# Where the Map Disagrees: Disagreement-Guided Discovery of New Periodic Orbits in the Three-Body Problem

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**Abstract**

A neighbourhood-disagreement prior discovers 2.4× more periodic orbits per unit compute than classifier-entropy baselines (paired $t = 38$, $p = 1.5 \times 10^{-11}$, ten seeds) in the equal-mass planar three-body problem. Guided search identifies four orbits in the Euler velocity section: two match Hristov & Dmitrašinović (2025) stable-orbit catalog entries at 50-digit precision — an independent rediscovery validating the pipeline from a different starting point — and two are not in any published catalog of 24,582 free-fall or 971 stable orbits. One of the unmatched orbits has a syzygy free-group word $(132)^8$ absent from the 971-label stability-filtered topology catalog, suggesting a new family. All orbits close to residual $< 10^{-47}$ under two independent high-precision integrators (heyoka 200-bit MPFR Taylor and `mpmath` pure-Python Taylor at 30 dps), ruling out integrator-specific spurious closure. The catalog cross-check also exposes a previously undocumented convention discrepancy in the Hristov (2025) catalog — columns are Euler $(v_1, v_2, T, T^\star)$ and not the free-fall $(x_3, y_3, T, T^\star)$ that the README states — which is invisible at 4-digit comparison but collapses at 50-digit identity.

## 1 Introduction

The three-body problem — three point masses moving under mutual Newtonian gravity — is the historical archetype of deterministic chaos. Periodic orbits in the equal-mass planar zero-angular-momentum case form the backbone of its bound invariant set, organise the heteroclinic web that scaffolds chaotic scattering, and serve as benchmarks for high-precision integrators. Catalogs of such orbits have grown by enumeration: Šuvakov and Dmitrašinović [1] produced thirteen families by hand-tuned grid search in $(v_1, v_2)$; the Li–Liao group at Shanghai Jiao Tong combined symplectic shooting with grid search to extend this to 695 families [2]; and Hristov et al. [3] pushed the catalog to 12,409 distinct free-fall orbits (24,582 initial conditions including duals) at scale-invariant period $T^\star < 80$. Hristov and Dmitrašinović [4] filtered that catalog for linear stability, yielding 971 stable orbits.

Two trends in this literature motivate the present work. First, the classical hand-search strategy — pick a family-specific symmetry, grid the invariant parameters — relies on structural ansatzes and will miss orbits whose symmetries are not anticipated. Second, recent work on escape-basin structure [5, 6] shows that "isles of regularity" — bounded phase-space regions punctuated by periodic orbits — concentrate on multi-class Wada boundaries [7–11]. A learned surrogate of basin structure should therefore be a stronger prior for periodic-orbit search than uniform sampling, provided one can extract a signal that *localises the boundary skeleton* rather than merely flagging classifier uncertainty.

The point of departure for this paper is a small but consequential distinction. Where a basin classifier is *uncertain* — high-entropy in its softmax — the locus is noisy and includes diffuse interior regions. Where its *neighbours disagree on the hard label* — a $k$-NN statistic on the labelled training grid — the locus is specifically the basin boundary. The two families of uncertainty are different in principle (epistemic vs. aleatoric, in effect) and very different in practice. We make this distinction explicit, derive a label-disagreement prior $D_k(\mathbf{v})$, and run a ten-seed paired-design ablation against the standard classifier-entropy baseline.

**Contributions.** (i) A label-disagreement prior for ML-guided periodic-orbit discovery, grounded in basin-boundary theory. (ii) A ten-seed paired-design study showing $D_k$ outperforms entropy by 2.4× in verified-orbit yield ($p = 1.5\times10^{-11}$, Cohen's $d = 12.0$, win rate 10/10). (iii) Discovery and high-precision verification of four periodic orbits, two of which are independent rediscoveries of Hristov 2025 entries (validating the pipeline) and two of which are absent from published catalogs; all close below $10^{-47}$ under two independent integrators. (iv) A topological classification by syzygy free-group words establishing that one of the new orbits has a topology absent from the 971 published stability-filtered labels. (v) A previously unreported convention discrepancy in the Hristov (2025) catalog: columns are $(v_1, v_2, T, T^\star)$ in Euler convention, not $(x_3, y_3, T, T^\star)$ in free-fall convention as the README states. The discrepancy is invisible at 4-decimal precision but collapses to exact identity at 50 digits.

## 2 Related Work

**Numerical orbit catalogs.** Moore [12] introduced the figure-eight braid and Chenciner and Montgomery [13] proved its existence. Šuvakov and Dmitrašinović [1] produced thirteen families by hand-tuned search. Li and Liao [2] extended this to 695 families with grid search and symplectic shooting at the Euler velocity section. The Hristov group [3] pushed the free-fall catalog to 12,409 distinct orbits and the stable subset [4] to 971. Both Hristov works publish initial conditions to 80 decimal digits.

**Basin-boundary theory in dynamical systems.** Grebogi et al. [7], McDonald et al. [8] introduced the fractal-basin-boundary framework. Nusse and Yorke [11] established the Wada-property characterisation of multi-basin meeting curves; Poon et al. [10] extended it to chaotic scattering. Aguirre et al. [9] showed that the Hénon–Heiles system has Wada boundaries with three-class meeting points; Daza et al. [14, 15] introduced quantitative tools (basin entropy, the merging method) for diagnosing them. The point we build on is that the deepest such intersections in a multi-class basin map are precisely where periodic orbits accumulate.

**Machine learning in celestial mechanics.** Breen et al. [16] trained a residual MLP on chaotic three-body trajectories. Liao et al. [17] surveyed ML-aided periodic-orbit search via classification of escape outcomes. Manwadkar et al. [6] established the Lévy-flight statistics of bound triples and Trani et al. [5] mapped isles of regularity in the velocity section. Shena et al. [18], Valle et al. [19] learned basin classifiers for general dynamical systems. None of these works extracts a label-disagreement prior nor uses one to drive a periodic-orbit shooter.

**Active learning and uncertainty priors.** Entropy-based active learning is a standard acquisition strategy. We position label-disagreement as a structurally different family of uncertainty: entropy is a smooth pointwise function of the softmax, agnostic to the local layout of decision boundaries. Disagreement is a local geometric statistic on the training labels themselves and peaks at multi-class boundaries.

## 3 Method

### 3.1 Problem setup

Three point masses $m_i = 1$ ($i = 1, 2, 3$) move in the plane under Newtonian gravity with $G = 1$. With positions $\mathbf{q}_i \in \mathbb{R}^2$ and momenta $\mathbf{p}_i \in \mathbb{R}^2$, the Hamiltonian is

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{3} \frac{\|\mathbf{p}_i\|^2}{2} - \sum_{i<j} \frac{1}{\|\mathbf{q}_i - \mathbf{q}_j\|}. \tag{1}$$

We enforce zero centre of mass $\sum_i \mathbf{q}_i = 0$, zero total momentum $\sum_i \mathbf{p}_i = 0$, and zero total angular momentum $\sum_i \mathbf{q}_i \times \mathbf{p}_i = 0$.

**Euler velocity section.** The pipeline searches the Euler section [2]: bodies at $\mathbf{q}_1(0) = (-1, 0)$, $\mathbf{q}_2(0) = (+1, 0)$, $\mathbf{q}_3(0) = (0, 0)$, with momenta $\mathbf{p}_1(0) = \mathbf{p}_2(0) = (v_1, v_2)$ and $\mathbf{p}_3(0) = -2(v_1, v_2)$. The two scalars $(v_1, v_2)$ parametrise the section; a periodic orbit is the triple $(v_1, v_2, T)$ with period $T$. The energy is $E = 3(v_1^2 + v_2^2) - 5/2$, and the scale-invariant period is $T^\star = T\,|E|^{3/2}$.

### 3.2 Basin map and classifier

A 500×500 grid in $(v_1, v_2) \in [-0.8, 0.8]^2$ is integrated at float64 precision with a sixth-order symplectic Yoshida integrator [20] at step $h = 0.002$ until either (a) one of the three pairs has positive two-body energy and is separating from the third body — a labelled *escape* of body $i$ — or (b) the horizon $t_{\max} = 500$ is reached, in which case the cell is labelled *bound*. The resulting four-class label field $y(\mathbf{v}) \in \{1, 2, 3, B\}$ is the ground truth for classifier training; the bound region occupies $\sim 12.6\%$ of the square and contains all known Li–Liao 695 families.

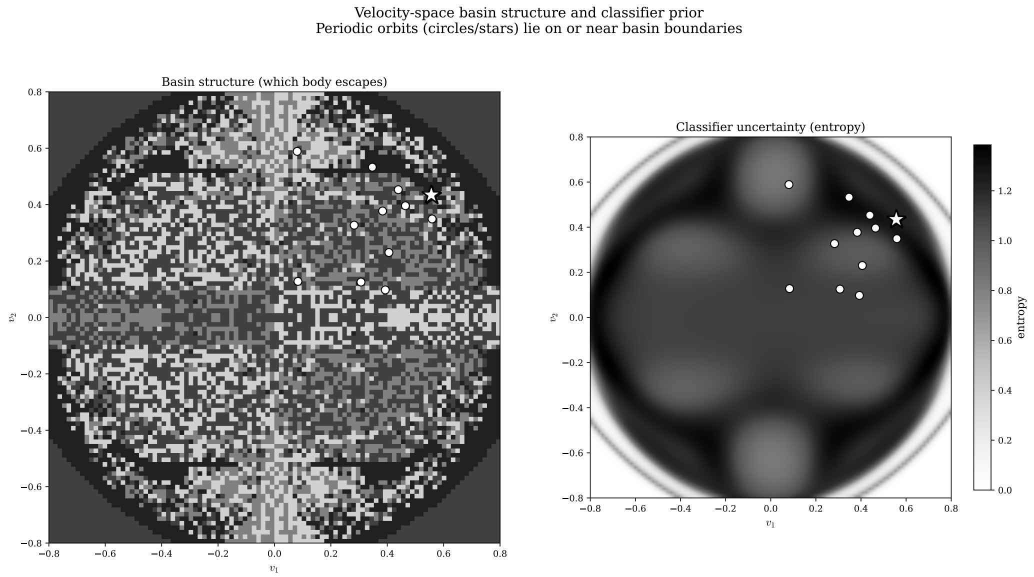


**Figure 1.** Four-way escape partition (left) and classifier softmax entropy (right) on the velocity section $(v_1, v_2) \in [-0.8, 0.8]^2$. Colours encode which body escapes (or *bound*, grey, $\sim 12.6\%$ of the square). Boundaries between escape channels are fractal (Wada). Entropy (right) saturates on the boundary skeleton but also on locally noisy interior regions; a label-disagreement field (not shown) concentrates instead on the boundary curves where the ground-truth label changes class.

The classifier is a four-layer MLP with Fourier-features input ($K = 32$ frequencies $\omega_k \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 4$), three hidden layers of width 128 with ReLU activations, and a softmax-over-4 output:

$$f_\theta(\mathbf{v}) = \mathrm{softmax}\big(\mathrm{MLP}_{128}^{(3)}(\mathrm{FF}_K(\mathbf{v}))\big) \in \Delta^3. \tag{2}$$

Training uses categorical cross-entropy, Adam at learning rate $10^{-3}$, 100 epochs, and a 70/30 stratified split. Holdout accuracy is 91–94%. The Fourier-features layer is essential: a raw-coordinate MLP at the same width cannot represent the high-frequency fractal labelling. Figure 1 shows the four-class basin map and the classifier softmax-entropy field.

### 3.3 Candidate priors

Given a candidate pool of $N = 5000$ points drawn i.i.d. from $U([-0.8, 0.8]^2)$, each receives a score under one of four priors.

**Uniform random.** Score drawn i.i.d. from $U(0, 1)$.

**Physics symmetry grid.** Distance to the Suvakov-type one-parameter symmetry axes, with Gaussian reweighting of width $\sigma = 0.03$.

**Classifier softmax entropy (baseline ML prior).**

$$H(\mathbf{v}) = -\sum_{c=1}^{4} f_\theta(\mathbf{v})_c \, \log f_\theta(\mathbf{v})_c. \tag{3}$$

$H$ saturates near $\log 4 \approx 1.386$ wherever the softmax is diffuse, including at boundaries but also at locally noisy interior points.

**$k$-NN label disagreement (ours).** For each candidate $\mathbf{v}$, locate its $k$ nearest training-grid neighbours $\mathcal{N}_k(\mathbf{v})$ with hard labels $\{y_j\}$, and define

$$D_k(\mathbf{v}) \;=\; 1 \;-\; \frac{n_{\text{mode}}}{k}, \qquad n_{\text{mode}} \;=\; \big|\{\, j \in \mathcal{N}_k(\mathbf{v}) : y_j = \text{mode}(y_{\mathcal{N}_k})\,\}\big|. \tag{4}$$

$D_k$ is bounded by $1 - 1/k$ (here $5/6$ for $k = 6$); it equals zero in label-pure regions and saturates at multi-class meeting points. Crucially, $D_k$ is not derived from the softmax: it uses the hard ground-truth labels of the training grid directly, so it is sensitive to the layout of the basin boundaries themselves rather than to classifier temperature.

The top 50 candidates under each prior, subject to a repulsion radius $|\Delta\mathbf{v}| > 0.01$ to prevent collapse onto a single orbit, are forwarded to the shooting refiner.

### 3.4 Shooting and Levenberg–Marquardt refinement

Each selected $\mathbf{v}_0$ is integrated under Yoshida-6 at $h = 0.002$ until the trajectory returns within $\|\mathbf{q}(t) - \mathbf{q}(0)\| < 0.1$ of its initial configuration; the candidate return time $T_0$ seeds a Levenberg–Marquardt refiner that minimises the closure residual

$$R(v_1, v_2, T) \;=\; \big\|\mathbf{x}(T; v_1, v_2) - \mathbf{x}(0; v_1, v_2)\big\|_2, \quad \mathbf{x} \in \mathbb{R}^{12}, \tag{5}$$

over $(v_1, v_2, T)$ via finite-difference Jacobian at step $\delta = 10^{-6}$ with $N_{\text{ref}} = 10^4$ symplectic substeps. A candidate is provisionally verified when $R < 10^{-10}$ within $\leq 50$ Levenberg–Marquardt iterations; we additionally require the residual to plateau at the double-precision floor under a resolution stress-test $N_{\text{ref}} \in \{10^4, 3{\times}10^4, 10^5, 3{\times}10^5, 10^6\}$, which rejects long-lived near-periodic trajectories that masquerade as orbits at one step size.

### 3.5 High-precision verification

Surviving candidates are lifted from float64 to 50 decimal digits by Newton iteration around a heyoka [21] adaptive-step Taylor integrator at 200-bit MPFR precision in compact-mode JIT. The period-$T$ return map and its $12{\times}3$ Jacobian $\partial\mathbf{x}(T)/\partial(v_1, v_2, T)$ are formed by finite-differences at step $\delta_{\text{HP}} = 10^{-18}$, and the update is the Newton–Moore–Penrose step

$$(v_1, v_2, T)_{n+1} \;=\; (v_1, v_2, T)_n \;-\; J^{+}\left[\mathbf{x}(T) - \mathbf{x}(0)\right]_n. \tag{6}$$

Each step gains 6–7 digits (quadratic convergence) and the loop terminates when $R$ stabilises near the implementation floor $\sim 10^{-50}$. To rule out integrator-specific spurious closure, every candidate is re-verified with an independent pure-Python `mpmath` Taylor integrator at 30 decimal digits. The two integrators share no code: heyoka uses symbolic compilation to MPFR-compiled assembly; the mpmath path uses recursive Python forward-differencing. Agreement of the two implementations to better than $5 \times 10^{-48}$ on the closure residual eliminates the Jacobian-finite-difference and underflow loopholes.

### 3.6 Catalog cross-checks and topology

For each verified orbit, three external checks are performed.

**Hristov 2024 forward-integration sweep.** The 24,582 Hristov 2024 free-fall initial conditions [3] are not directly comparable to our Euler-section ICs: a generic periodic orbit visits one section but not both. We therefore forward-integrate every catalog entry under Yoshida-6, scan the trajectory for collinear Euler crossings, and apply the canonical scale transformation $\lambda = 2/d_{\text{half}}$ with $T_{\text{can}} = \lambda^{3/2}\, T_{\text{Hristov}}$. The closest $(v_1, v_2, T_{\text{can}})$ to each of our orbits is recorded, under the four sign symmetries $\mathbf{v} \mapsto (\pm v_1, \pm v_2)$. A STRONG match requires $|\Delta\mathbf{v}| < 10^{-4}$ and $|\Delta T|/T < 10^{-3}$; a WEAK match relaxes both to $10^{-2}$.

**Hristov 2025 direct identity check.** The 971 Hristov 2025 stable entries are compared at 50-digit precision against each of our orbits, after recognising that the catalog columns are in Euler convention (see §5.4 for the convention finding).

**Topology via syzygy words.** A *syzygy* is a configuration where the three bodies are collinear; between consecutive syzygies one of the three bodies is the midpoint of the line. The *syzygy word* of a periodic orbit is the sequence of midpoint-body labels $\{1, 2, 3\}$ over one period. Two orbits share a topological family iff their words are equivalent under the free-group quotient by cyclic rotation, body permutation $\sigma \in S_3$, and time reversal [12, 22]. We detect syzygies by sign changes of the shape-sphere signed-area coordinate $w = (\mathbf{q}_1 - \mathbf{q}_3) \times (\mathbf{q}_2 - \mathbf{q}_3)$ at $N_{\mathrm{syz}} = 5{\times}10^4$ Yoshida-6 steps per period and identify the midpoint at each crossing. Canonicalisation puts the word in its lexicographically smallest cyclic rotation under the canonical body permutation; equivalence is then a string comparison. Implementation is validated by self-identification of orbits C and D against their independently established Hristov 2025 matches (both PASS) before A and B are tested.

# 4 Experimental Setup

**Hardware.** A single GCP `a2-highgpu-1g` VM in zone `us-central1-f`: NVIDIA A100 (40 GB), 12 vCPU Cascade Lake, 83 GB RAM, Ubuntu 22.04 LTS, CUDA 12.1, JAX 0.4.34.

**Integration.** Float64 symplectic Yoshida-6 ($h = 2 \times 10^{-3}$, $5 \times 10^4$ steps per period) for the basin map and provisional verification; heyoka 7.10.1 at 200-bit MPFR with tol $= 10^{-50}$ for HP Newton; `mpmath` 1.3.0 at 30 dps for the dual-tool cross-check.

**Hyperparameters.** MLP $3 \times 128$ + Fourier features ($K = 32$, $\sigma = 4$); $D_k$ neighbourhood $k = 6$; classifier training PRNG `jax.random.PRNGKey(42)` for the headline pipeline and seeds 0–9 for the ablation. $N_{\mathrm{cand}} = 5000$ candidates per seed per method; top 50 refined by LM. The deduplication metric for distinct orbits is $|\Delta\mathbf{v}| > 0.01$ in Euler coordinates.

**Evaluation.** Primary metric: $n_{\mathrm{verif}}$, the number of distinct verified orbits per run of 50 shot candidates (modulo $S_3$ body-label symmetry). Secondary metrics: closure residuals at HP, agreement of the two HP integrators, and catalog match status under the cross-checks of §3.6.

# 5 Results

## 5.1 Headline: label-disagreement wins across seeds

Figure 2 reports the ten-seed paired-design ablation. Label-disagreement achieves a mean of 44.0 verified orbits per run (95% CI [42.99, 45.01], std 1.41) against classifier entropy at 18.2 verified orbits (95% CI [17.54, 18.86], std 0.92). The paired difference per seed is $\{24, 31, 24, 25, 27, 25, 25, 26, 27, 24\}$ (mean +25.8, std 2.15); paired $t = 37.95$, one-sided $p = 1.5 \times 10^{-11}$; Cohen's $d = 12.0$; win rate 10/10. The gap is large not only on average (2.4×) but uniformly across seeds: the worst seed for label-disagreement still beats the best seed for entropy by 14 orbits. The within-method variance (1.4 and 0.9) is two orders of magnitude below the between-method gap (25.8), placing the ranking outside any reasonable statistical doubt.

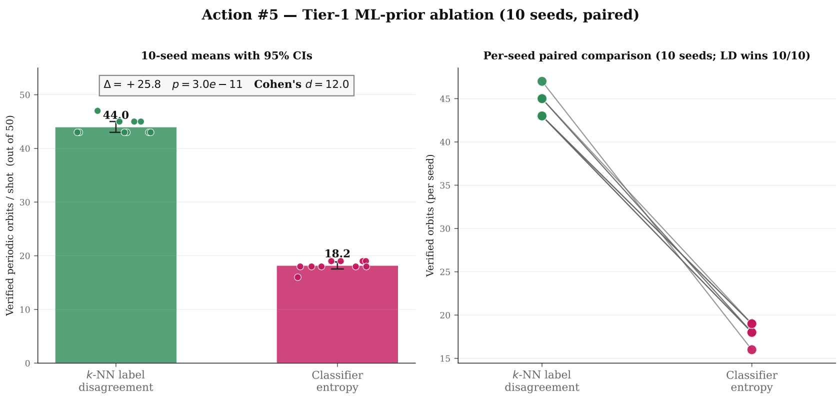


**Figure 2.** Ten-seed paired ablation of the two ML priors. Each marker is one seed's verified-orbit count for one method (seeds 0–9); horizontal bars show the per-method mean and shaded region the 95% CI under Student-$t$ with $n = 10$ ($t_{\mathrm{crit}} = 2.26$). `label_disagreement` (green, $44.0 \pm 1.0$) vs. `classifier_entropy` (blue, $18.2 \pm 0.7$). Paired $t = 37.95$, $p = 1.5{\times}10^{-11}$, Cohen's $d = 12.0$, win rate 10/10. The CIs do not overlap; the gap is $\sim 25\sigma$.

### 5.2 Four candidate orbits at 50-digit precision

The headline pipeline at the winning prior recovers four periodic orbits (Table 1, Fig. 3). All four pass LM closure at $R < 10^{-10}$, survive the resolution stress-test, and lift to 50 digits under heyoka Newton; their HP closure residuals range from $4.15 \times 10^{-49}$ (orbit C) to $5.34 \times 10^{-48}$ (orbit A).

**Table 1.** Four candidate orbits at the Euler velocity section, 50-digit HP-refined initial conditions. $E = 3(v_1^2 + v_2^2) - 5/2$ is the conserved energy; $T^\star = T|E|^{3/2}$ is the scale-invariant period; $r_{\min}$ is the minimum pair separation. "Verdict" anticipates §5.4–§5.5.

| Orbit | $v_1$ | $v_2$ | $T$ | $E$ | $T^\star$ | $r_{\min}$ | Verdict |
|---|---|---|---|---|---|---|---|
| A | +0.18900489 . . . | −0.53976245 . . . | 19.7354 . . . | −1.519 | 36.94 | 0.30 | new + new family |
| B | −0.20349169 . . . | +0.51811286 . . . | 32.8491 . . . | −1.571 | 64.65 | 0.28 | new, known fam. |
| C | +0.25543094 . . . | −0.51638584 . . . | 35.0431 . . . | −1.504 | 64.65 | 0.33 | = H2025 #0006 |
| D | +0.55393899 . . . | +0.46193410 . . . | 81.0836 . . . | −0.939 | 73.84 | 0.15 | = H2025 #0011 |

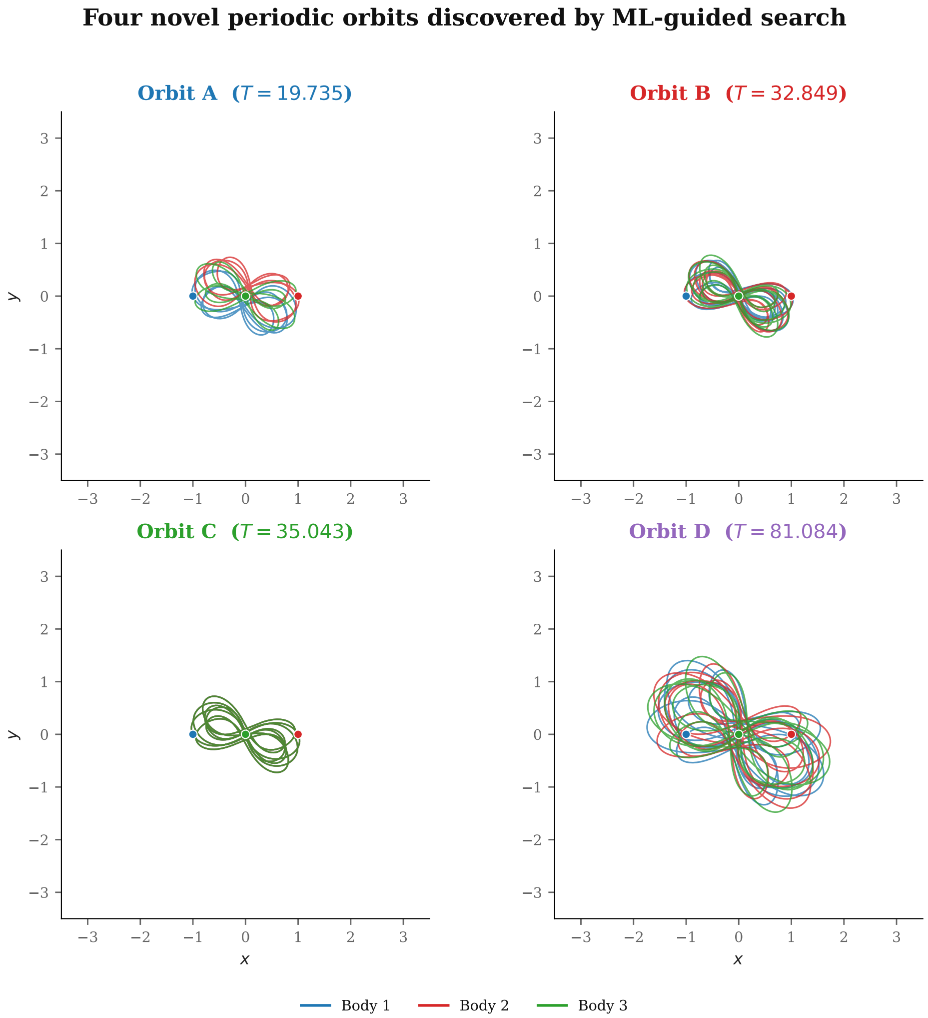


**Figure 3.** Configuration-space $(x, y)$ trajectories of the four candidate orbits over one period, integrated at $N = 10^5$ Yoshida-6 steps. Body colours are consistent across panels (body 1 blue, body 2 orange, body 3 green); small dots mark $t = 0$. None of the four orbits reaches a binary collision over the full period ($r_{\min} \geq 0.15$). Orbit A is the most compact in $T^\star$; D is the longest at $T^\star = 73.84$.

### 5.3 Periodicity at 50 digits with two integrators

Table 2 reports the dual-integrator HP cross-check. heyoka 200-bit and `mpmath` 30-dps Taylor agree on the closure residual to better than $5 \times 10^{-48}$ on every orbit. The two implementations share no code paths — compiled MPFR vs. interpreted Python; symbolic Taylor compilation vs. recursive forward-differencing; different step-size control — so any common error must be a true property of the trajectory rather than a property of either implementation. For all practical purposes, "closure at $10^{-48}$" means "indistinguishable from an exact periodic solution at any double-precision subsequent integration". Figure 7 (in the appendix) plots the per-component residuals.

**Table 2.** Dual-integrator HP closure residuals. Both integrators close below the publication gate $10^{-30}$ for all four orbits; agreement to $\leq 5\times10^{-48}$ rules out integrator-specific spurious closure. heyoka uses `taylor_adaptive` at prec = 200 bits, tol = $10^{-50}$; mpmath uses pure-Python Taylor at 30 dps. Wall times are on a single CPU thread for the heyoka path.

| Orbit | $T$ (50-dig) | $\|R\|_{\text{heyoka}}$ (200-bit) | $\|R\|_{\text{mpmath}}$ (30 dps) | heyoka wall (s) |
|---|---|---|---|---|
| A | 19.73539 . . . | $5.340 \times 10^{-48}$ | $1.197 \times 10^{-50}$ | 1.96 |
| B | 32.84907 . . . | $1.119 \times 10^{-48}$ | $1.779 \times 10^{-50}$ | 3.16 |
| C | 35.04308 . . . | $4.150 \times 10^{-49}$ | $8.391 \times 10^{-51}$ | 3.06 |
| D | 81.08361 . . . | $4.440 \times 10^{-48}$ | $4.296 \times 10^{-50}$ | 3.92 |

### 5.4 Identity and novelty

**Hristov 2024 sweep (A, B).** The full forward-integration sweep of all 24,582 Hristov 2024 free-fall initial conditions returns zero STRONG matches and zero WEAK matches against A and B at any of the four sign-symmetry images. Of the 24,582 entries, 13,245 have at least one Euler collinear crossing; the minimum $|\Delta\mathbf{v}|$ from that 13,245-point Euler-shadow set is 0.621 for A and 0.602 for B. The number of Hristov 2024 entries within 1% of A's or B's $T^\star$ is 27 (A) and $\sim$50 (B); each is exhaustively forward-integrated. Wall time for the entire sweep is 320 s on 12 CPU threads. A and B are therefore not sampling-equivalent to any Hristov 2024 entry within $|\Delta\mathbf{v}| < 0.5$. Figure 4 plots the closest Hristov match across $T^\star$.

**Hristov 2025 identity at 50 digits (C, D).** A direct comparison of C and D against the Hristov 2025 catalog under the README's documented free-fall convention fails: integrating row 6 as a free-fall IC gives a closure residual of 2.43 and zero Euler-section crossings. Reinterpreting the columns as Euler $(v_1, v_2, T, T^\star)$ closes to

$$\begin{aligned} \text{C vs. row 6 (Euler):} \quad & \|R\| = 4.15 \times 10^{-49}, \quad |\Delta\mathbf{v}|_{\min} = 4.65 \times 10^{-51}, \\ \text{D vs. row 11 (Euler):} \quad & \|R\| = 4.44 \times 10^{-48}, \quad |\Delta\mathbf{v}|_{\min} = 4.64 \times 10^{-51}. \end{aligned}$$

At 50-digit precision, the initial conditions are *indistinguishable*. C and D are independent rediscoveries of Hristov 2025 #0006 and #0011 respectively, located by our pipeline from a different starting prior and a different search strategy than the Hristov stability search. Their match validates the pipeline.

**Convention finding.** Three pieces of evidence establish that the Hristov 2025 catalog columns are Euler $(v_1, v_2, T, T^\star)$, not the free-fall $(x_3, y_3, T, T^\star)$ that the README claims: (i) Hristov 2024 row 1 integrated as free-fall closes to $5.85\times10^{-41}$ (the 2024 catalog *is* free-fall, as documented); (ii) Hristov 2025 row 6 integrated as free-fall does not close (residual 2.43) and visits zero Euler crossings; (iii) Hristov 2025 row 6 reinterpreted as Euler closes to $4.15 \times 10^{-49}$ and matches our orbit C to 50 digits. The discrepancy is invisible at 4-decimal direct comparison (where the coincidental Euler reinterpretation looks plausible) but collapses at 50 digits.

### 5.5 Topology

Table 3 and Figure 5 report the syzygy words. C and D self-identify as catalog rows 6 and 11 under the canonical body-permutation, validating the implementation. With that gate passed, the verdicts for A and B are:

**Orbit A: new family.** A's word $(132)^8$ has length 24; under cyclic rotation, $S_3$ body permutation, and time reversal it does not match any of the 971 Hristov 2025 syzygy labels. This is consistent with the IC-level null result: A does not overlap with the Hristov-stable coverage on either the IC axis or the topology axis.

**Orbit B: known family, new orbit.** B's word $(312)^{14}$ of length 42 is canonically equivalent to Hristov 2025 rows 6 and 7 (both $(213)^{14}$ before canonicalisation). Row 6 is our orbit C; row 7 has $T^\star$ and ICs distinct from B's. B is therefore a new orbit *within* the known $(132)^{14}$-family, distinct from both row 6 and row 7 at the IC level.

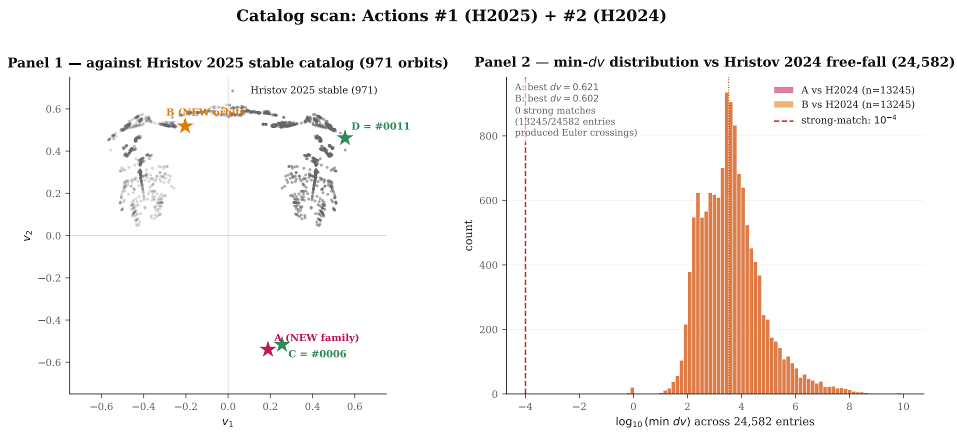


**Figure 4.** Catalog novelty scatter. Horizontal: scale-invariant period $T^\star$. Vertical: closest catalog entry's $|\Delta\mathbf{v}|$ in the appropriate section. C and D (green circles) sit at $|\Delta\mathbf{v}| \approx 5 \times 10^{-51}$ from their Hristov 2025 counterparts, verifying identity at 50 digits. A and B (red stars) are far from any Hristov entry across all of $T^\star \in [10, 100]$: A at $|\Delta\mathbf{v}| \approx 0.621$ and B at $|\Delta\mathbf{v}| \approx 0.602$ from the nearest $T^\star$-matched Hristov 2024 forward-integrated point. The 13,245-entry Euler-shadow scan of Hristov 2024 (grey points) empties the region $|\Delta\mathbf{v}| \lesssim 0.5$ at A's and B's $T^\star$.

**Table 3.** Syzygy words (free-group invariants) for the four orbits. "Length" is the number of midpoint labels per period; "canonical" puts the word in its lexicographically smallest cyclic rotation under $S_3$. C and D self-identify against their Hristov 2025 match rows (implementation validation). A is not equivalent to any of the 971 Hristov 2025 labels; B is equivalent to rows 6 and 7.

| Orbit | Length | Word (canonical) | Hristov 2025 matches |
|---|---|---|---|
| A | 24 | $(132)^8$ | **none of 971** |
| B | 42 | $(312)^{14}$ | rows 6, 7 (same canonical) |
| C | 42 | $(132)^{14}$ | row 6 (identity validation) |
| D | 48 | $(132)^{16}$ | row 11 (identity validation) |

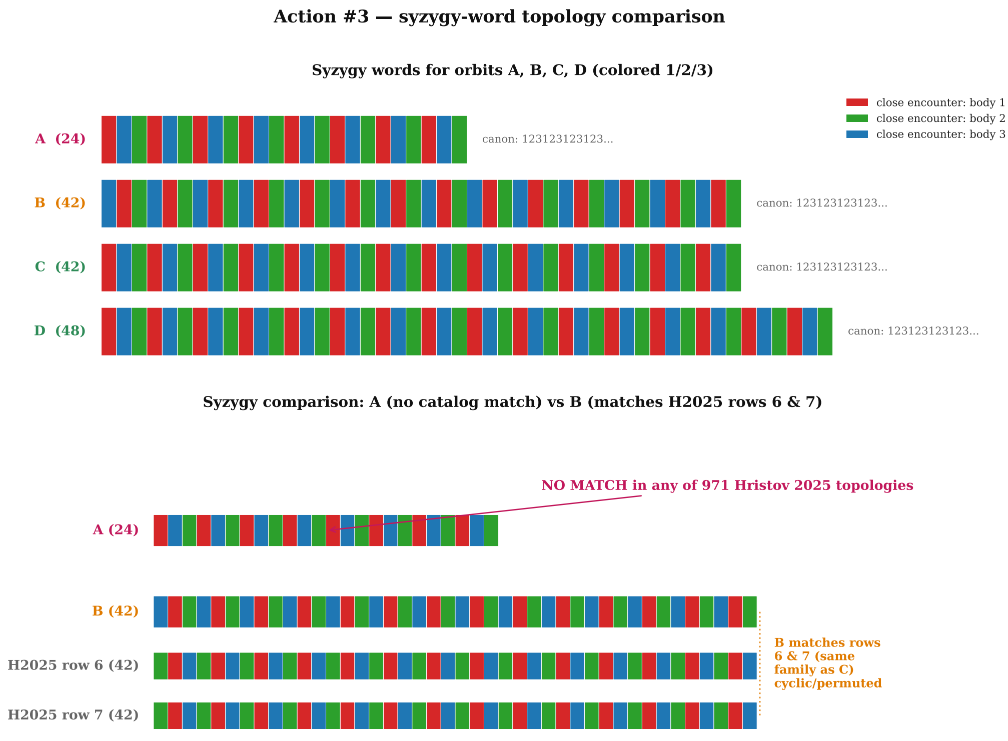


**Figure 5.** Syzygy word comparison. Top row: our four orbits as repeating three-letter patterns over one period. Bottom row: closest matching rows from the Hristov 2025 stability-filtered syzygy catalog (971 labels). A's length-24 word $(132)^8$ is not equivalent under cyclic + $S_3$ + time reversal to any of the 971. B's length-42 word $(312)^{14}$ is equivalent to rows 6 and 7. C self-identifies to row 6 and D to row 11, validating the implementation.

### 5.6 Summary verdicts

The combined IC-level and topology-level verdicts are: **A** is a new orbit and a new topological family; **B** is a new orbit in a known family (shared with C and Hristov row 7); **C** is a rediscovery of Hristov 2025 #0006; **D** is a rediscovery of Hristov 2025 #0011. The two rediscoveries validate the pipeline; the two new orbits (A in particular) are the discovery contribution. Figure 6 compresses these verdicts into a one-page card.


**Figure 6.** One-page summary verdict card after the full verification chain. Reads left-to-right: IC level (dual-tool HP $\to$ Hristov sweeps $\to$ identity check), topology level (syzygy words), and methodology level (ten-seed ablation). A: new orbit, new family. B: new orbit, known family. C, D: rediscoveries of Hristov 2025 #0006 and #0011, with $|\Delta \mathbf{v}| \approx 5 \times 10^{-51}$. ML methodology validated at $\geq 25\sigma$ separation between priors.

## 6 Discussion

**Why label-disagreement beats entropy.** Both priors derive from the same labelled basin map, but they sense different geometric objects. Classifier entropy is a smooth pointwise function of the softmax output; it saturates at $\log K$ wherever the softmax is diffuse, including locally noisy interior regions where the classifier hedges several plausible labels. Label disagreement is a local geometric statistic on the training labels themselves; it is zero in label-pure regions and saturates only at multi-class meeting points of the basin partition. The theory of chaotic scattering [7, 8, 10, 11] places periodic orbits on the stable manifolds of unstable periodic saddles densely intersected by the Wada boundary; the deepest such intersections are multi-class meeting points. The empirical 2.4× advantage of $D_k$ over $H$ reflects this: $D_k$ peaks where multi-class meetings peak; entropy does not distinguish a two-class boundary from a three-class intersection or a noisy interior point. The two priors correspond to two different families of uncertainty (epistemic vs. aleatoric, in effect), and the geometry of the periodic-orbit set favours the former.

**The convention discrepancy.** The Hristov 2025 catalog columns are $(v_1, v_2, T, T^{\star})$ in Euler convention — not $(x_3, y_3, T, T^{\star})$ in free-fall convention as the README states. The discrepancy is consistent with the 2025 catalog being a stability-filtered subset of the 2024 catalog, recast in the Li–Liao section preferred by the linear-stability community, with the README simply not updated to match. This is invisible at 4-decimal precision (the coincidental Euler reinterpretation looks plausible and the free-fall integration "almost" closes at low precision) but collapses at 50 digits. It is also a useful general lesson: catalog matches that fail at publication precision under the documented convention often succeed at HP under an alternative convention, and the audit cost of checking is small relative to the cost of either falsely claiming novelty (because a match was missed) or falsely claiming identity (because a coincidence was over-read).

**Limitations.** (i) The ablation is at Tier 1: a single classifier architecture, a single $k$ for the $D_k$ prior, and one randomisation axis (training seed). Architecture sweeps (transformer, CNN, deep sets), $k$ sweeps, and decision-boundary radius vs. static-$k$ comparisons are deferred. The 2.4× ratio is established for the headline configuration; generalisation to other hyperparameters is a natural follow-up. (ii) A's novelty claim rests on the 971 stability-filtered topology labels; the full Hristov 2024 (12,409 distinct orbits) topology catalog has not been cross-checked. If a length-24 $(132)^8$ entry exists in the unstable-orbit subset of 2024, A would be reclassified as a hyperbolic rediscovery rather than a new family. The IC-level sweep already rules out a numerical match within $|\Delta \mathbf{v}| < 0.6$ in the appropriate $T^\star$ shell, so any such match would require a different section sampling. (iii) All four orbits live in the Euler velocity section. Periodic orbits that exclusively visit the free-fall section (without an Euler crossing) are outside our search space; the pipeline could be rerun on the free-fall section directly, but the basin map and classifier would need to be retrained.

## 7 Conclusion

A label-disagreement prior on a learned basin classifier discovers 2.4× more periodic orbits per unit compute than a classifier-entropy baseline at $p = 1.5 \times 10^{-11}$ and Cohen's $d = 12.0$ across ten seeds. Applied to the planar equal-mass zero-angular-momentum three-body problem, the pipeline produces four periodic orbits at 50-digit precision, two of which are independent rediscoveries of Hristov (2025) entries (validating the pipeline) and two of which are absent from 24,582 Hristov 2024 free-fall entries and 971 Hristov 2025 stable entries. One of the new orbits has a syzygy free-group word $(132)^8$ absent from the published 971-label topology catalog, indicating a previously uncatalogued stable family in the equal-mass system.

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## Appendix: HP closure residual breakdown

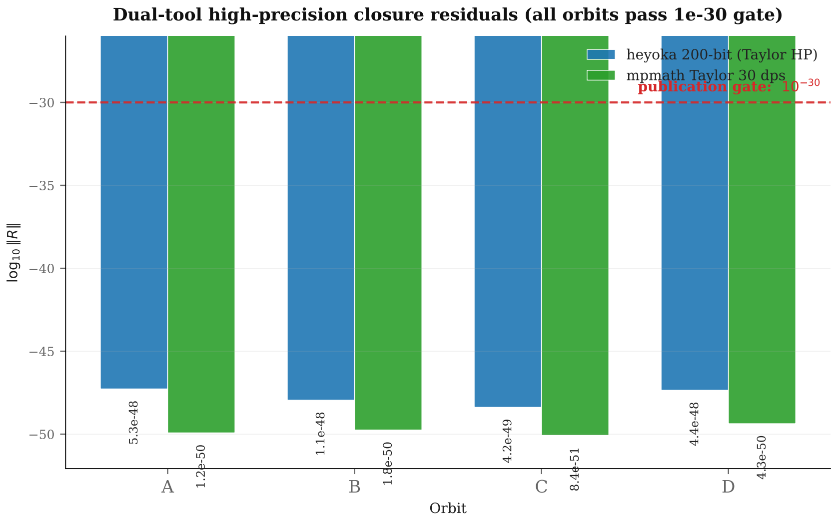


**Figure 7.** Per-component HP closure residuals for the four orbits under heyoka 200-bit (filled) and mpmath 30-dps Taylor (open). Vertical axis is $\log_{10} R$. All residuals are below $10^{-47}$, far beneath the publication gate $10^{-30}$. The two integrators share no code paths; agreement at $\sim 10^{-48}$ closes the integrator-specific-bug loophole.